

# Slide layout gallery

A worked example of every demolab slide layout.

Demolab · SLIDES.md D11

# Bullets

- **Bold the load-bearing phrase** so the eye lands on it first.
- Two lines per bullet, maximum — past that it's prose, not a slide.
- Five bullets is the ceiling; split the slide before a sixth.
- Nested points are fine **once**, for a short aside.
- The last bullet is the **so what** — say what it means.

# Two-column

## What a tool does

- Holds reusable computation.
- Writes `numbers.json` + data.
- Stays language-agnostic.

## What a tool does not

- Render plots.
- Import a runner.
- Hard-code a result.

**Parallel sides, one takeaway: tools compute, runners narrate.**

# Three-column

## Tools

Reusable computation;  
emit data, not plots.

## Experiments

Runners: call the tools,  
render the figures.

## Writings

One `.typ` per entry;  
reads the run.

**Three parallel items — four wants a table.**

# Code panel

```
def cmd_lif(args):  
    v, spikes = simulate(args.current, args.tau_m)  
    write_output(run_dir,  
        {"firing_rate_hz": rate(spikes)},  
        manifest={"headline_metrics": ["firing_rate_hz"]})
```

## Equation + terms

$$\tau_m \frac{dV}{dt} = -(V - V_{\text{rest}}) + R_m I_{\text{ext}}(t)$$

where:

- $V$  — membrane potential (mV)
- $\tau_m$  — membrane time constant (ms)
- $V_{\text{rest}}$  — resting potential (mV)
- $R_m$  — membrane resistance ( $\text{M}\Omega$ )
- $I_{\text{ext}}(t)$  — external input current (nA); the space keeps  $(t)$  out of the subscript (D10)

# Quote

*“An article about computational science is not the scholarship itself; it is merely advertising of it. The actual scholarship is the complete code and data that generated the figures.”*

— Buckheit & Donoho, 1995

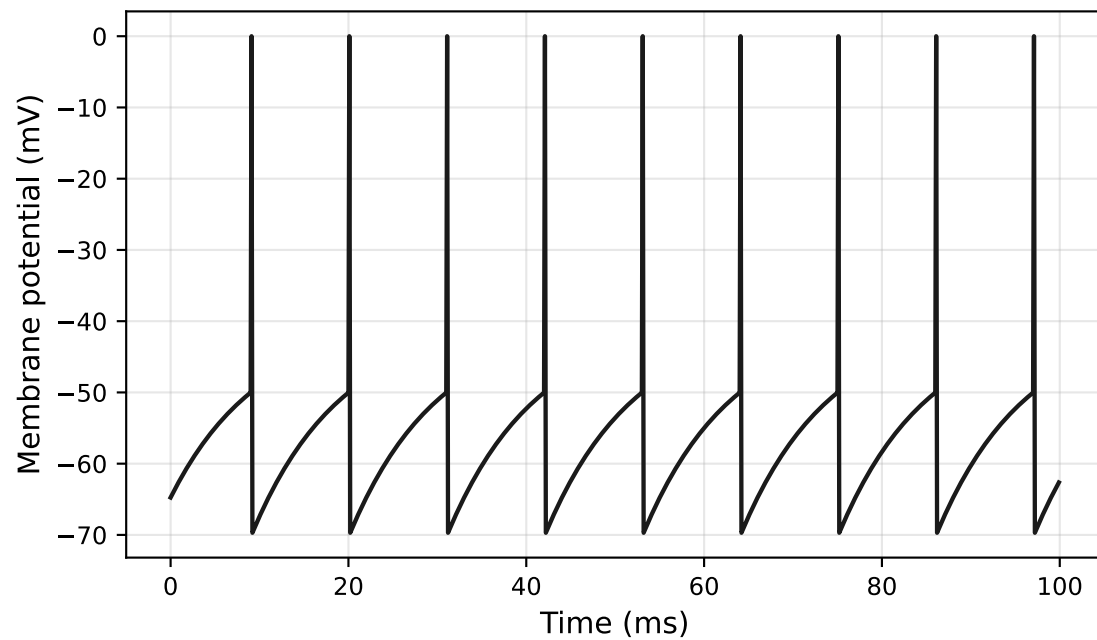
PART TWO

# Figures

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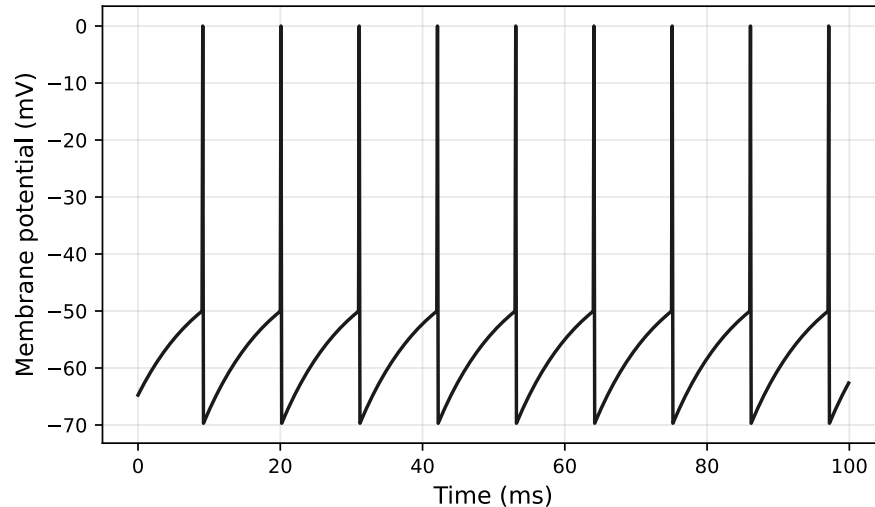


# Centered figure



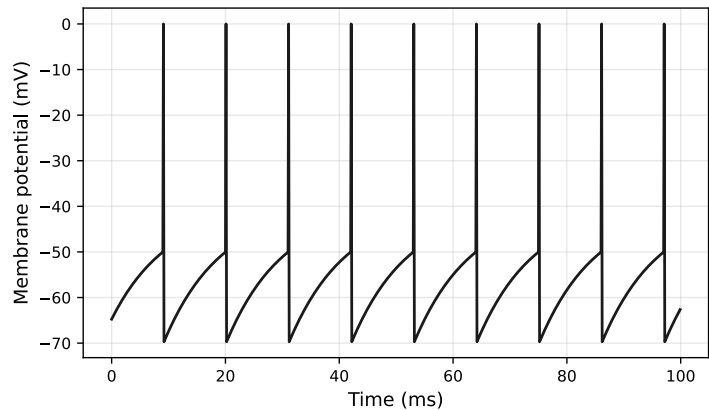
Membrane potential of a single LIF neuron under tonic input; it charges and resets at threshold, firing at 90 Hz.

## Figure + bullets

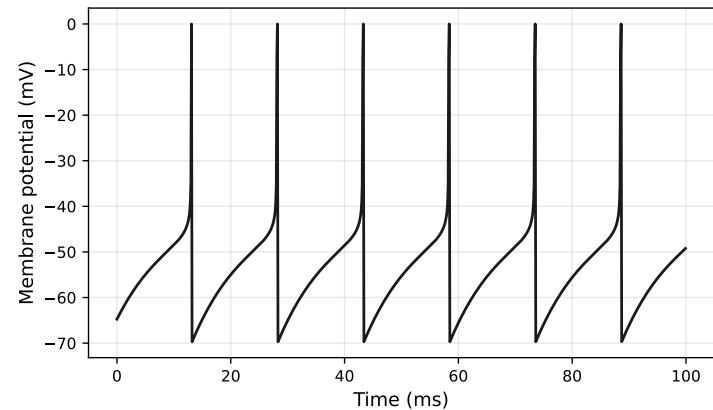


- Each sweep is the membrane charging through its RC constant.
- Every vertical drop is a **reset** after a spike.
- **So what:** constant drive gives a fixed rate, 90 Hz.

# Figure pair



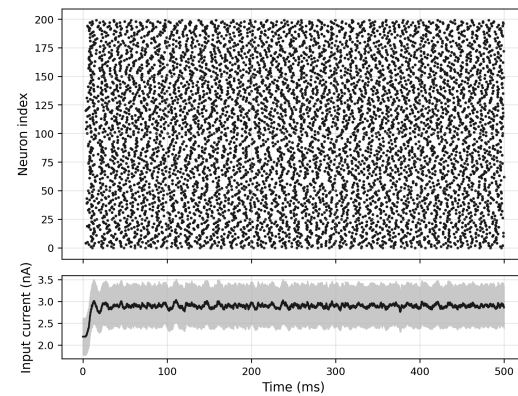
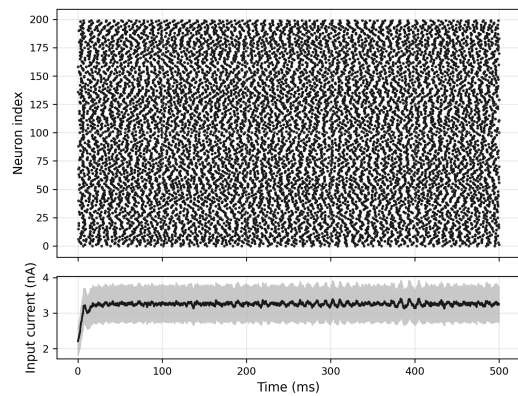
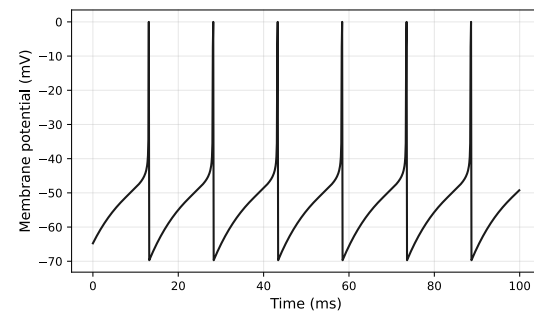
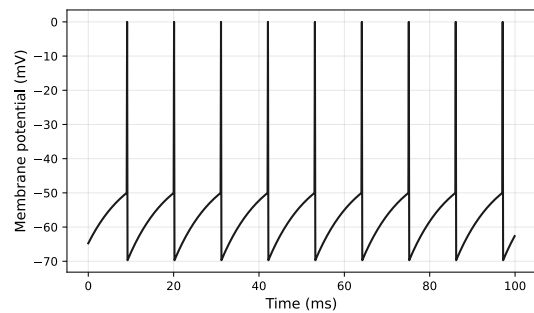
(a) LIF: linear integrate-and-fire.



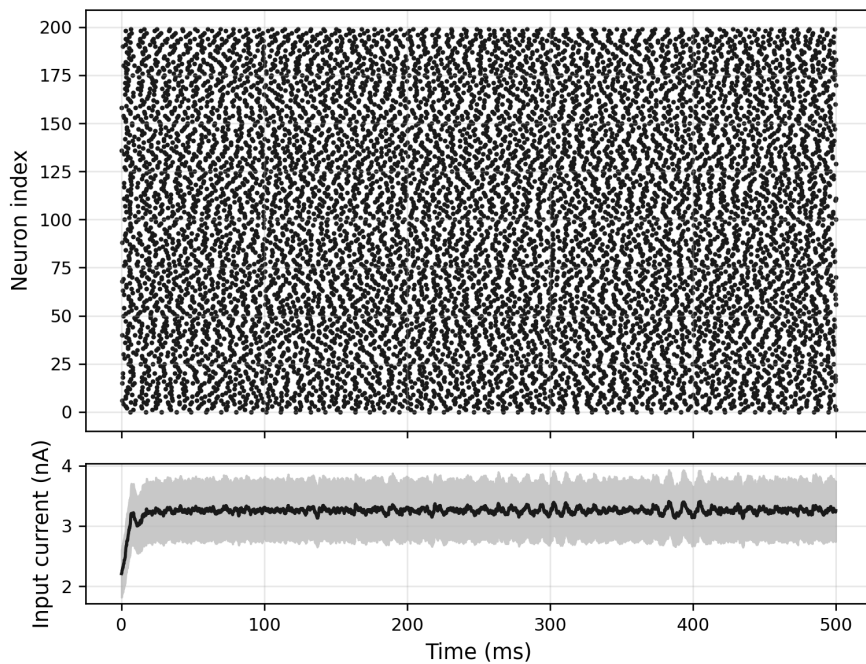
(b) EIF: exponential spike onset.

**Same drive, sharper threshold: the EIF spike initiates faster.**

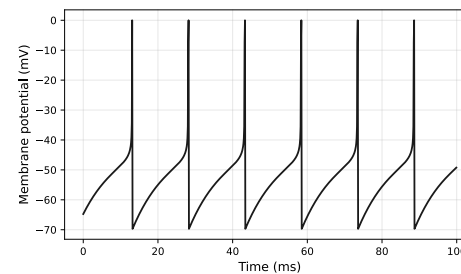
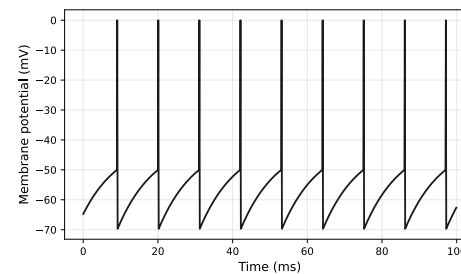
# Figure grid (2×2)



# Hero + stack

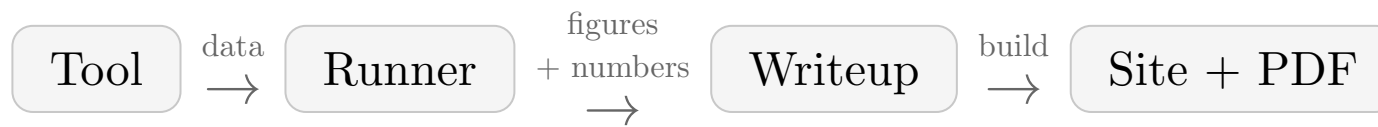


The network sustains irregular firing — the result being argued.



Single-cell traces as supporting evidence.

# Diagram



**Boxes and arrows in Typst — no image asset needed.**

# Table

| Parameter                       | Value  |
|---------------------------------|--------|
| Input current                   | 2.5 nA |
| Membrane time constant $\tau_m$ | 10 ms  |
| Threshold $V_{th}$              | −50 mV |
| Firing rate                     | 90 Hz  |

Every value read from `numbers.json`, so the slide can't drift from the run.

Big number

90 Hz

single-neuron firing rate under tonic input — read from the run



One layout per slide.

Title, plus bullets or one visual.

# Closer

## demolab-engine/guides/SLIDES.md

- Copy a layout from this gallery; don't re-derive it.
- Check the page count after every edit (D9) — overflow paginates silently.
- Numbers and figures come from the run, never the keyboard.

Every layout, one demolab look.